

Limits:

$$\lim_{x \rightarrow +\infty} \left[\ln(4x^2+1) - \ln(3x^2+7) \right]$$

$$\Rightarrow \ln\left(\frac{4x^2+1}{3x^2+7}\right)$$

$$\Rightarrow \lim_{x \rightarrow +\infty} \ln\left(\frac{4x^2+1}{3x^2+7}\right) = \frac{4x^2+1}{3x^2+7} = \frac{\infty}{\infty} \text{ I.S.}$$

L'Hôpital Rule:

$$\frac{d}{dx}(4x^2+1) = 8x \quad \Rightarrow \quad \frac{8x}{6x} = \frac{8}{6} = \frac{4}{3}$$

$$\frac{d}{dx}(3x^2+7) = 6x$$

$$\text{So: } \ln\left(\frac{4}{3}\right)$$

$$\lim_{x \rightarrow +\infty} \left[\ln(4x^2+1) - \ln(3x^2+7) \right] = \ln\left(\frac{4}{3}\right)$$



$$\lim_{x \rightarrow 3} \frac{\ln(3x-1) - 3\ln 2}{(2x-5)^4 - 1}$$

$$\Rightarrow \frac{\ln(3(3)-1) - 3\ln 2}{(2(3)-5)^4 - 1} = \frac{\ln 8 - \ln 8}{(1)^4 - 1} = \frac{0}{0} \text{ I.f}$$

L'Hôpital Rule:

$$\frac{d}{dx} (\ln(3x-1) - 3\ln 2) = \frac{3}{3x-1} - \frac{0}{8} = \frac{3}{3x-1}$$

$$\frac{d}{dx} ((2x-5)^4 - 1) = 4(2x-5)^3 \cdot 2 = 8(2x-5)^3$$

$$\Rightarrow \frac{\frac{3}{3x-1}}{8(2x-5)^3} = \frac{3}{(3x-1)(8)(2x-5)^3}$$

$$\text{So, } \frac{3}{64}$$

$$\lim_{x \rightarrow 3} \frac{\ln(3x-1) - 3\ln 2}{(2x-5)^4 - 1} = \frac{3}{64}$$

[Signature]